

Group 6

DAY: _____

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⇒ Past Paper Fall 2023

Question # 01:-

Answer the following short questions.

• ~~(i) 3~~ •

Multivibrator:-

Multivibrator is an electronic circuit that oscillates between two or more distinct states, creating continuous output signals.

- It is constructed by connecting two amplifying transistors or valves, each with its output to the input of others, it produces oscillatory signals.

OR

In simple:-

A multivibrator is an electronic circuit that generates square rectangular, pulse waveform also called called as non-linear oscillator or function generator.

• Application •

Excess - 3 code :-

Excess - 3 code is binary-coded decimal (BCD) encoding scheme, where each decimal digit is represented by its binary equivalent plus 3

Example:

1. Decimal digit 2 :-

Binary equivalent : 0010

Excess - 3 code : $0010 + 0011 = 0101$

2. Decimal digit 5 :-

Binary equivalent: 0101

Excess 3 code : $0101 + 0011 = 1000$

→ The binary scheme is used in most digital systems and networks. It is called as 'series-2's' because each binary code is more than the actual binary representation of the decimal digit.

Ex iii B

Convert $(001101010)_2$ to hexadecimal.

Sol Solution B

→ Firstly;

Group the binary digits into 4-bit chunks from right to left;

• 0011 1010

→ Secondly;

convert each group into its hexadecimal equivalent

• $0011 = 3$

• $1010 = A$

So,

00111010 (binary) = 3A (hexadecimal)

• Qiv B

Convert $(798)_8$ to binary
octal

In ~~decimal~~ system;

$$7 = 0111$$

$$9 = 1001$$

$$8 = 1000$$

So

$$(798)_8 = (011110011000)_2$$

• Qv B

Is NOR Gate associative
or not?

The NOR gate is not associative.
Associativity means that the
grouping of input doesn't affect
the final output, but for
NOR gates, the order of
operation can change the
results.

~ (vi) ~

Convert $(01101011101)_2$
into Gray code!

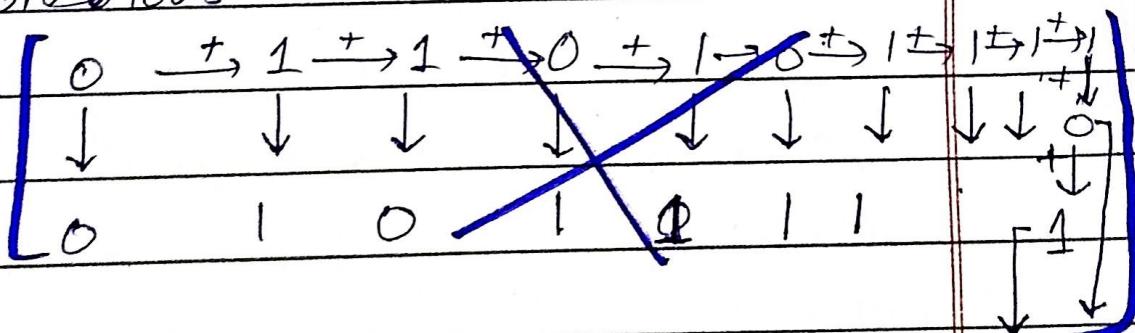
Firstly;

The first bit of the Gray code is similar to the first bit of the binary code

$$\text{Gray } \{0\} = \text{Binary } \{0\} = 0$$

Secondly;

Each subsequent Gray bit is obtained by Xoring the current binary bit with the previous



$$\begin{array}{cccccccccccc}
 0 & + & 1 & + & 1 & + & 0 & + & 1 & + & 0 & + & 1 & + & 1 & + & 1 & + & 1 & + & 0 & + & 1 \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 0 & & 1 & & 0 & & 1 & & 1 & & 1 & & 0 & & 0 & & 0 & & 1 & & 1 & & 1
 \end{array}$$

So;

$$(01101011101)_2 = (010111100011)_{\text{Gray}}$$

~ (vii) ~

Differentiate between amplifier and oscillator

Amplifier	Oscillators
→ Boost the strength of an input signal without changing its frequency	→ Generate a repetitive signal without needing an input signal.
→ The output of an amplifier is a stronger version of input signals.	→ The output of an oscillator is a continuous periodic waveform
→ Amplifier uses negative feedback to stabilize gain.	→ Oscillator use positive feedback to maintain oscillations.

Uses

Audio system
Radio receivers

Uses

Signal Generator
RF transmitters.

~(viii)~

Name the two types of oscillators:-

Oscillators:-

An oscillator is an electronic circuit that produces a repetitive signal, often used in electronic devices to generate a stable frequency. It converts DC power into AC power, creating a continuous waveform.

Types:-

- Harmonic oscillators (produces sinusoidal signal)
- Relaxation oscillators (produces non-sinusoidal signal, i.e. square waveform)

~(ix)~

How does class AB differ from class B Amplifier?

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In class B amplifier, both transistors are biased at cut-off where as in class AB amplifier, both transistors are biased slightly above cut off.

~~Q. Ex 10~~

Convert the given expression into standard form:

$$X = AB + ABD + BCD$$

For Simplifying

$$X = AB + ABD + BCD$$

Factor out common terms

$$X = AB(1 + D) + BCD$$

$\therefore 1 + D = 1$ by Law 2 of OR law

$$X = AB(1) + BCD$$

$$X = B(A + CD)$$

~(xi)~.

What are two types of clipping in Class A Amplifier?

In class A amplifier, clipping occurs when the ~~input~~ output signal exceeds the amplifier's linear operating range.

Types:-

1. Peak Clipping

2. Saturation Clipping

Both type of clipping can cause distortion and degradation of the output signals.

~(xii)~.

Bandwidth:-

A range of frequency within a given band, in particular that used for transmitting a signal.

Formula:-

$$B.W = f_2 - f_1$$

• Q(xiii) B •

Miller Effect:-

A phenomenon where apparent input capacitance of an amplifier or device is increased due to the presence of capacitance between its input and output terminals.

• Q(xiv) B •

Differentiate between inverting and non-inverting configuration of operational Amplifier?